

## NOTES: CLASS-4

2. Obtain the Fourier series for  $f(x) = \begin{cases} -\pi, & -\pi < x < 0 \\ x, & 0 < x < \pi \end{cases}$

Hence deduce that  $\frac{\pi^2}{8} = \frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots$

**Solution :** The Fourier expansion of  $f(x)$  over  $(-\pi, \pi)$  is

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$

where

$$\begin{aligned} a_0 &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx \\ &= \frac{1}{\pi} \left[ \int_{-\pi}^0 -\pi dx + \int_0^{\pi} x dx \right] \\ &= \frac{1}{\pi} \left\{ [-\pi(x)]_{-\pi}^0 + \left[ \frac{x^2}{2} \right]_0^{\pi} \right\} \\ &= \frac{1}{\pi} \left[ -\pi^2 + \frac{\pi^2}{2} \right] \end{aligned}$$

$$\Rightarrow a_0 = -\frac{\pi}{2}$$

$$\begin{aligned} a_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx dx \\ &= \frac{1}{\pi} \left[ \int_{-\pi}^0 f(x) \cos nx dx + \int_0^{\pi} f(x) \cos nx dx \right] + \frac{1}{\pi} \left[ \int_{-\pi}^0 -\pi \cos nx dx + \int_0^{\pi} x \cos nx dx \right] \\ &= \frac{1}{\pi} \left[ -\pi \left[ \frac{\sin nx}{n} \right]_{-\pi}^0 + \left[ x \cdot \frac{\sin nx}{n} \right]_0^{\pi} - \left[ -\frac{\cos nx}{n^2} \right]_0^{\pi} \right] \\ &= \frac{1}{\pi(n^2)} \left[ \cos nx \right]_0^{\pi} \\ &= \frac{1}{\pi n^2} [\cos n\pi - \cos n(0)] \end{aligned}$$

$$\Rightarrow a_n = \frac{1}{\pi n^2} [(-1)^n - 1]$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx dx$$

$$\begin{aligned}
 &= \frac{1}{\pi} \left[ \int_{-\pi}^0 f(x) \sin nx \, dx + \int_0^{\pi} \square(x) \sin nx \, dx \right] + \frac{1}{\pi} \left[ \int_{-\pi}^0 -\pi \sin nx \, dx + \int_0^{\pi} x \sin nx \, dx \right] \\
 &= \frac{1}{\pi} \left[ -\pi \left[ \frac{-\cos nx}{n} \right]_{-\pi}^0 + \left[ x \cdot \frac{-\cos nx}{n} \right]_0^{\pi} - \left[ -\frac{\sin nx}{n^2} \right]_0^{\pi} \right] \\
 &= \frac{1}{\pi(n)} \left\{ \left[ \pi \cos nx \right]_{-\pi}^0 - \left[ x \cos nx \right]_0^{\pi} \right\} \\
 &= \frac{1}{\pi n^2} \left[ \pi \cos n(0) - \pi \cos n\pi - [\pi \cos n(\pi) - 0] \right] \\
 \Rightarrow b_n &= \frac{1}{n} [1 - 2(-1)^n]
 \end{aligned}$$

The required Fourier series is given by

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$

$$\begin{aligned}
 f(x) &= \frac{1}{2} \left( \frac{-\pi}{2} \right) + \sum_{n=1}^{\infty} \left\{ \frac{1}{\pi n^2} [(-1)^n - 1] \cos nx + \frac{1}{n} [1 - 2(-1)^n] \sin nx \right\} \\
 \Rightarrow f(x) &= \frac{-\pi}{4} + \sum_{n=1}^{\infty} \left\{ \frac{1}{\pi n^2} [(-1)^n - 1] \cos nx + \frac{1}{n} [1 - 2(-1)^n] \sin nx \right\}
 \end{aligned}$$

**Deduction :**

To deduce the series we shall put  $x = 0$  in the above equation.

$$f(0) = -\frac{\pi}{4} + \sum_{n=1}^{\infty} \left\{ \frac{1}{\pi n^2} [(-1)^n - 1] \cos n(0) \right\}$$

Since  $x = 0$  is a point of discontinuity,

$$\begin{aligned}
 f(0) &= \frac{1}{2} [f(0-0) + f(0+0)] \\
 &= \frac{1}{2} [-\pi + 0] \\
 &= -\frac{\pi}{2}
 \end{aligned}$$

Therefore,

$$\frac{-\pi}{2} = \frac{-\pi}{4} + \frac{-1}{\pi} \sum_{n=1}^{\infty} \left( \frac{1}{n^2} [1 - (-1)^n] \right)$$

$$\Rightarrow \frac{\pi}{2} = \frac{\pi}{4} + \frac{1}{\pi} \left( \frac{2}{1^2} + \frac{2}{3^2} + \frac{2}{5^2} + \dots \right)$$

$$\Rightarrow \frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots = \frac{\pi^2}{8}$$

2. Express  $f(x) = (\pi - x)^2$  as a Fourier Series of period  $2\pi$  in the interval  $0 < x < 2\pi$  and hence deduce the sum of the series  $1 + \frac{1}{2^2} + \frac{1}{3^2} + \dots$

**Solution:** The Fourier expansion of  $f(x)$  over  $(0, 2\pi)$  is

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$

where

$$\begin{aligned} a_0 &= \frac{1}{\pi} \int_0^{2\pi} f(x) dx. \\ &= \frac{1}{\pi} \int_0^{2\pi} (\pi - x)^2 dx \\ &= \frac{1}{\pi} \left[ \frac{(\pi - x)^3}{3(-1)} \right]_0^{2\pi} \\ &= \frac{-1}{\pi} \left[ \frac{-\pi^3}{3} - \frac{\pi^3}{3} \right] \end{aligned}$$

$$\Rightarrow a_0 = \frac{2\pi^2}{3}$$

$$\begin{aligned} a_n &= \frac{1}{\pi} \int_0^{2\pi} f(x) \cos nx dx \\ &= \frac{1}{\pi} \int_0^{2\pi} (\pi - x)^2 \cos nx dx \\ &= \frac{1}{\pi} \left[ (\pi - x)^2 \left( \frac{\sin nx}{n} \right) - 2(\pi - x)(-1) \left( -\frac{\cos nx}{n^2} \right) + 2(-1) \left( -\frac{\sin nx}{n^3} \right) \right]_0^{2\pi} \\ &= \frac{1}{\pi} \left[ 2(\pi - x) \left( -\frac{\cos nx}{n^2} \right) \right]_0^{2\pi} \\ &= \frac{1}{\pi n^2} [ +2\pi(\cos 2n\pi) + 2\pi \cos n(0) ] \\ &= \frac{1}{\pi n^2} (4\pi) \end{aligned}$$

$$\Rightarrow a_n = \frac{4}{n^2}$$

$$b_n = \frac{1}{\pi} \int_0^{2\pi} f(x) \sin nx dx$$

$$\begin{aligned}
 &= \frac{1}{\pi} \int_0^{2\pi} (\pi - x)^2 \sin nx \, dx \\
 &= \frac{1}{\pi} \left[ (\pi - x)^2 \left( \frac{-\cos nx}{n} \right) - 2(\pi - x)(-1) \left( -\frac{\sin nx}{n^2} \right) + 2(+1) \left( \frac{\cos nx}{n^3} \right) \right]_0^{2\pi} \\
 &= \frac{1}{\pi} \left[ (\pi - x)^2 \left( \frac{-\cos nx}{n} \right) + \frac{2}{n^3} (\cos nx) \right]_0^{2\pi}
 \end{aligned}$$

$$\Rightarrow b_n = 0$$

The required Fourier series is given by

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$

$$\Rightarrow (\pi - x)^2 = \frac{2\pi^2}{3} \left( \frac{1}{2} \right) + \sum_{n=1}^{\infty} \left( \frac{4}{n^2} \cos nx + (0) \sin nx \right)$$

$$\Rightarrow (\pi - x)^2 = \frac{\pi^2}{3} + \sum_{n=1}^{\infty} \left( \frac{4}{n^2} \cos nx \right)$$

**Deduction :**

To deduce the series we shall put  $x = 0$  in the above the series

$$f(0) = \frac{\pi^2}{3} + \sum_{n=1}^{\infty} \left( \frac{4}{n^2} \cos n(0) \right)$$

Since,  $x = 0$  is a point of discontinuity,

$$f(0) = \frac{1}{2} [f(0 - 0) + f(0 + 0)] = \frac{1}{2} [\pi^2 + \pi^2] = \pi^2$$

$$\text{Therefore, } f(0) = \pi^2 = \frac{\pi^2}{3} + \sum_{n=1}^{\infty} \left( \frac{4}{n^2} \right)$$

$$\Rightarrow \frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \cdots \cdots \cdots = \frac{\pi^2}{6}$$

**REFERENCES:**

- 1. Higher Engineering Mathematics-B.S. Grewal**
- 2. Higher Engineering Mathematics-B.V. Ramana**

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